

Liu Cheng

Shanghai, China | +86-15975314558

liucheng20021023@gmail.com | github.com/liucheng532 | [LiuchengHomePage](#)

Education

The Hong Kong University of Science and Technology (Guangzhou)

Guangzhou, China

M.Phil. in Robotics and Autonomous Systems

Sep. 2024 – Oct. 2026

- Awarded a **Full M.Phil. Scholarship**.

Nanjing University of Posts and Telecommunications

Nanjing, China

B.Eng. in Electromagnetic Fields and Wireless Technology

Sep. 2020 – Jun. 2024

- Awarded the **National Encouragement Scholarship** (Top 5%) and **First-Class Academic Scholarship** (Top 5%).

Experience

OpenDriveLab, The University of Hong Kong

Hong Kong

Research Assistant | Humanoid Robotics

Sep. 2026 – Present

- Contributed to **PingPongDuo: Agile Humanoid Table Tennis Doubles with Safe Team Coordination (ICRA2027)**, leading the **RL planner sim-to-real integration** and building the deployment pipeline from motion capture, robot state estimation, and coordinate calibration to real-world execution on humanoid robots.
- Currently working on **GPT-6-driven humanoid locomotion-manipulation** and **humanoid soccer** systems.

RoboCup 2026 World Champion | Humanoid Soccer Control

Mar. 2026 – Present

M.Phil. Thesis Project | Humanoid Motion Control

HKUST-GZ

- Developed a unified **recovery-locomotion** policy for the Booster T1 humanoid using PPO and AMP, enabling the robot to autonomously recover from falls and resume locomotion with a single controller; successfully achieved **sim-to-real deployment** on the physical robot.
- Built a **residual self-play** framework on top of a pretrained humanoid dribbling policy, improving possession survival and win rate in multi-robot competitive scenarios; successfully achieved **sim-to-real deployment** on the physical Booster T1.
- The project has reached a **publication-ready stage**, and the humanoid soccer work is being consolidated into a research manuscript.

ATEC2025 Real-World Extreme Challenge

Sep. 2025 – Dec. 2025

Team Leader | Core Developer

CUHK & Alibab

- Led team to **8th Place** in a national challenge on legged locomotion and precision manipulation.
- Developed a **teleoperation pipeline** for robotic arms Data collection and trained walking strategies for bimanual robots using RL. ([GitHub Link](#))

IROS 2025 Robot Dance Competition

Jun. 2025 – Oct. 2025

Project Leader

Guangdong, China

- Developed a **Video-to-Dance pipeline** for Unitree G1 humanoid robot via motion retargeting and Whole-Body Control (WBC); won **3rd Place**. ([GitHub Link](#))

THU AIR Embodied AI Winter Camp

Jan. 2025 – Feb. 2025

Outstanding Camper (Top Tier)

Tsinghua University AIR, Beijing

- Built a simulation data collection pipeline and deployed **ACT (Action Chunking with Transformers)** model for imitation learning on robotic arm manipulation tasks.

Future dynamics (Startup) — Co-Founder

Mar. 2024 – Present

Technical Lead

HKUST(GZ), Guangdong

- Lead technical strategy and executive management for the startup's robotics hardware-software integration projects.

Xbotpark Entrepreneurial Camp

Jul. 2022 – Aug. 2022

Team Leader

Changzhou, Jiangsu

- Developed a medical IoT demo using STM32, RFID, and WiFi modules to optimize patient hospital workflows; awarded **Best Project**.

DJI Robomaster | Teaching Assistant

Feb. 2022

Internship

DJI, Shenzhen

- Managed and maintained lab equipment and campus computing infrastructure for the Robomaster summer camp.

Robocup 3D Simulation Competition

Nov. 2020 – Nov. 2022

Team Leader | Algorithm Developer

NJUPT

- Implemented bipedal robot role assignment and collaboration using **Hungarian/KM algorithms**.
- Optimized bipedal walking actions via **CMA-ES** based on a linear inverted pendulum model (LIPM).

Publications

PingPongDuo: Agile Humanoid Table Tennis Doubles with Safe Team Coordination

Sep. 2026

- Haoran Jiang*, Yichao Zhong, Kai Zhang, **Cheng Liu**, Yixuan Pan, Zhenjie Yang, Jingbo Wang, Tianyu Li, Li Chen. *Manuscript under review*.

Towards Exploratory and Focused Manipulation with Bimanual Active Perception: A New Problem, Benchmark and Strategy

Feb. 2026

- Yuxin He, Ruihao Zhang, Tianao Shen, **Cheng Liu**, Qiang Nie. *IEEE International Conference on Robotics and Automation (ICRA), 2026*. [Accepted]

Projects

Information Bracelet for the Medical Internet (Embedded Design, Communication)

Jul.2022 – Aug.2022

Robocup 3D Simulation Competition (Planning, Decision, Optimization)

May.2021 – Nov.2022

Autonomous Tracking Car (Computer Vision, Embedded System)

Nov.2020 – Feb.2021

Personal Drivable FTC/Robomaster Robot (Manufacture, mechanical design)

Sep.2018 – Nov.2019

Technical Skills

Languages: Python, C/C++, ROS, Isaac Gym, Mujoco, PyTorch, Mjlab

Algorithms: K-means, CMA-ES, PID, Matching algorithm, DRL

Developer Tools: Git, Docker, Linux (Ubuntu)

Hardware: CAD (Solidworks), Embedded System(stm32, C51, Arduino), Electronic design(PCB)

Honors

RoboCup 2026 Soccer 3D Simulation League World Champion

Jul. 2026

HKUST(GZ) Campus Top 10 Singing Competition Champion

Jun. 2026

ATEC2025 · Real-World Extreme Challenge 8th Place

Dec. 2025

The 2025 IROS Mecha Challenge (Robot Dance) 3rd Place

Oct. 2025

Honor degree in EE, General Scholarship (Top 5%), NJUPT

May. 2024

BYD Campus New Voice Singing Competition Divisional Champion

Oct. 2023

National Inspiration Scholarship

Sep. 2023

HKUST(GZ) Red Bird Challenge Camp Winning Prize

Jul. 2023

Academic Merit Scholarship (First Prize)

Sep. 2022

Xbotpark 2022 Innovation Summer Camp Best Project

Aug. 2022

2022 RoboCup China Competition National Championship

Jul. 2022

2021 Jiangsu Robot Competition Provincial First Prize

Nov. 2021

2021 RoboCup WorldCup Third Prize

May. 2021

2019 DJI RoboMaster Robot Summer Camp Third Prize

Jul. 2019



Cheng Liu

湖北 黄石



扫一扫上面的二维码图案，加我为朋友。